

PART 2

SUPERPOSITION YIELDS ROBUST NEURAL SCALING

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WORDS (3000)

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Abstract

Neural scaling laws have demonstrated that bigger models can be expected to show greater performance in deep learning, but the theoretical explanation behind it is incomplete. This project is inspired by the paper Superposition Yields Robust Neural Scaling presented at NeurIPS 2025, which introduces the framework of representation superposition, and an orthogonal regularisation strategy to boost the separation of features. A synthetic sparse dataset, and superposition autoencoder were implemented and tested extensively with varying hidden dimensions in PyTorch. Results were able to capture the power-law scaling behaviour with significant decrease in feature overlaps, negligible reconstruction loss.

1. Introduction

Neural scaling laws have shown that, for deep learning models, bigger models, more training data and more computation have consistently led to an increase in performance [1]. These scaling behaviours are crucial for the development of more effective, resilient and scalable AI systems. These scaling laws have been extensively documented but there is a lack of understanding of the theories behind them. Because of representations learning, a neural network is able to learn meaningful features in the latent space and because of representation relationships are typically expressed as a power law as in the above form with (L) representing the loss and (N) representing size/computational budget, (A) and (C) as constants, and (alpha) as the scaling exponent. These relationships can be exploited to optimise the use of computational resources, and can also help in developing scalable deep learning systems.

$$L(N) = AN^{-\alpha} + C$$

A theoretical explanation for these scaling behaviours is that neural networks represent much more than the number of available dimensions and that the representation superposition is realized via overlap in the representations. Also, multiple features can be used in the same latent space and they are linearly separated from one another [2]. This mechanism can increase the representational efficiency, although low dimensional, and this is how modern neural networks are able to use high dimensional embeddings successfully.

One of the papers chosen for NeurIPS 2025 wonders if this robustness of neural scaling is an explanation of representation superposition. They systematically vary the latent dimensions and weight decay, and with the help of their theoretical toy model with sparse feature representations, they investigate the superposition, multiple features can be represented using the same representation, which represents much more than the network itself. One such paper, Superposition Yields Robust Neural Scaling from NeurIPS 2025 offers a theoretical explanation for this phenomenon and could serve as a starting point for exploring this correlation between superposition and neural scaling [1].

This project aims to tackle the lack of a more in-depth explanation of the theoretical origins of these behaviours, since current studies of neural scaling laws have tended to be descriptive. The enhanced understanding can be leveraged to create more efficient and explainable deep

learning models. With this in mind, the aim of this project is to replicate and extend the superposition of representations paradigm from the selected paper, and to explore the effects on the scaling behaviour of the neural model.

2. Background and Literature Review

Scaling laws for neural network models are well-established patterns of model performance as a function of the size of the model, the size of the data set and the amount of compute needed [2]. Parameter scaling: increases the representational capability of the network, data scaling: exposes the model to more patterns, and compute scaling: allows larger scale optimisation during training. These transition between the weak and strong superposition regime. Through their experiments, they find that the scaling behaviour is stable when strong superposition is used, agreeing with the observations of large language models in power-law [3]. The most outstanding strengths of the paper are the thorough theoretical basis and the effective application of the results to various experimental conditions; on the other hand, the experimental setup is quite simple and the results are not straightforward to apply to other settings.

Complementary work on sparse auto encoders features learning, and mechanistic interpretability and neural scaling. In the research on sparse autoencoders, efficient disentanglement of the latent features is shown; in the research on mechanistic interpretability, the internal features of neural networks are explored; and research on scaling laws mainly characterises the empirical performance trends [4]. However, this selected paper applies both representation learning and scaling theory, while most of these papers only work on one or the other of these concepts. But theoretical explanations that can be well validated under different architectures and experimental settings and computational constraints are missing [5]. Also, very little research has been done about adaptive regularisation techniques which may enhance the representation efficiency without compromising the scaling behaviour. The gaps between the theoretical, the experimental, the computational and the practical aspects led to the necessity of the implementation and evaluation of an enhanced representation superposition framework in this project.

3. Methodology

3.1 Overall Workflow

The general experimental procedure was put together to try to replicate and build upon the representation superposition idea introduced in the target NeurIPS paper in a systematic way. This was done by constructing a synthetic data set with a handful of features that are drawn from a power law frequency distribution to allow a controlled study of the superposition behaviour. The created data were then fed into a baseline superposition autoencoder, which modelled the original experimental setup. Then a modified autoencoder with an orthogonal regularisation is proposed which will ensure more efficient feature representation and less overlap between the latent features.

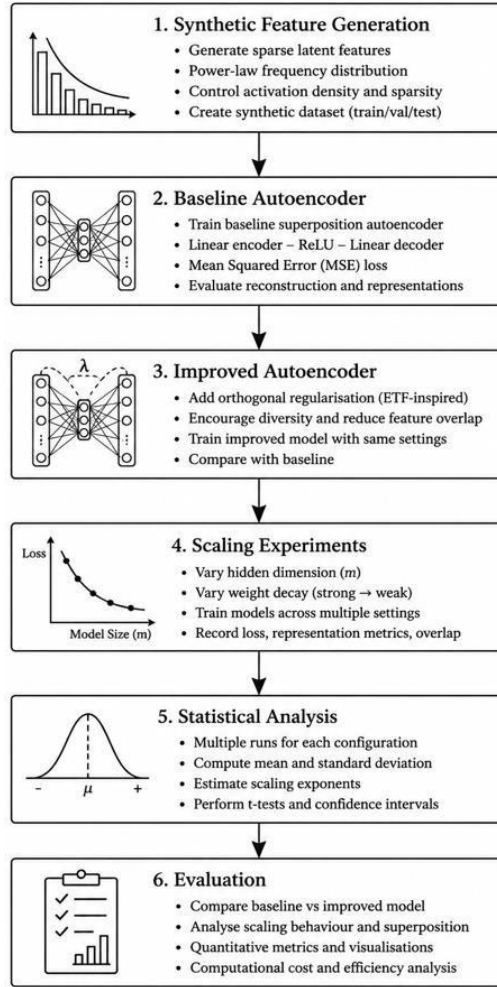


Figure 1: Workflow Diagram

The outputs were statistically analysed and repetition of experiments were carried out along with significance testing and exponent of scaling. Finally, quantitative evaluation metrics and computational cost analysis and comparative visualisations were used to evaluate the effectiveness, robustness and scalability of the proposed improvement over the baseline implementation.

3.2 Development Environment

The work has been implemented in Python using PyTorch deep learning framework on google colab. If available, to save training time GPU was used, otherwise CPU was used. Processing and visualisation of data was also carried out with the help of the libraries such as NumPy, Pandas, Matplotlib, Seaborn, SciPy, and TQDM for keeping track of progress. Random seeds are fixed; training is performed using automatic mixed precision (AMP) to enhance the speed of training while maintaining consistent numbers and deterministic algorithms are used to ensure reproducibility.

3.3 Dataset

A synthetic dataset was created which closely resembles the experimental setup used in the selected paper, with the aim of studying representation superposition in a systematic way, without the variability of real-world datasets. Sparse latent features are sampled from each sample using a sampling method that is a power-law random sampling of the sampling frequency distribution, or an exponential or uniform random sampling of the frequency distribution. For this study, a power-law distribution was chosen, because it is a known distribution that models the unequal frequencies of occurrence that are common in natural data sets. To maintain sparsity and the same number of expected active features/sample, the activation probabilities of the features were normalised using a fixed activation density [6]. The magnitudes of the features were normally distributed and were multiplied with binary activation masks to get the final data set.

3.4 Baseline Superposition Autoencoder

The baseline model was a lightweight autoencoder that was able to replicate the representation superposition framework mentioned in the selected paper. The architecture consisted of an input layer to which the synthetic feature vectors were input and a linear encoder which mapped the input to a lower-dimensional latent space. Before a linear decoder was used to recreate the original input, a ReLU activation function was used to add non-linearity [7]. The parameters of the model were initialised with Xavier uniform initialisation and biases of the decoder were set to zero to favour stable optimisation of the model.

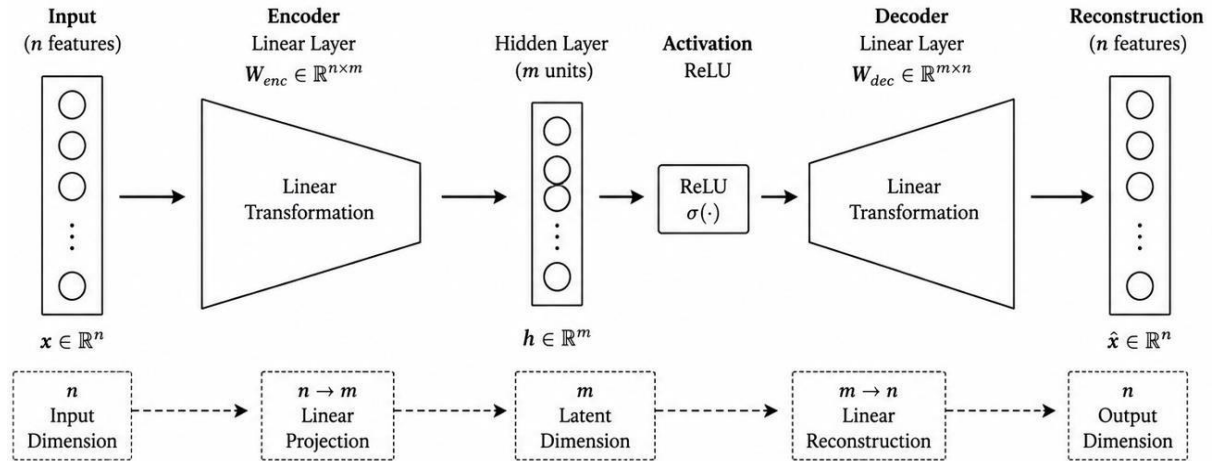


Figure 2: Baseline Superposition Autoencoder Architecture

Mean Squared Error (MSE) reconstruction loss was minimised during training with the use of the AdamW optimiser. During forward propagation, the input was successively passed to the encoder, ReLU activation and passed to the decoder, and the reconstruction error was calculated to update the weights in training. The simple structure is illustrated in figure 2.

3.5 Proposed Improvement

An orthogonal regularisation mechanism in the encoder was added to improve the model; geometry of Equiangular Tight Frame (ETF) is used for this purpose. Unlike the original implementation which only learns representations of the features by the reconstruction loss, the proposed approach attempts to explicitly minimize the correlation between the weight vectors

of the encoder, thus minimizing the overlapping of features in latent space. This was performed by normalising the weight vectors of the encoders, and forming a Gram Matrix of pairwise cosine similarities for all feature representations. Self-similarity is shown by the diagonal elements and the amount of overlap between features is shown by the off diagonal elements [8].

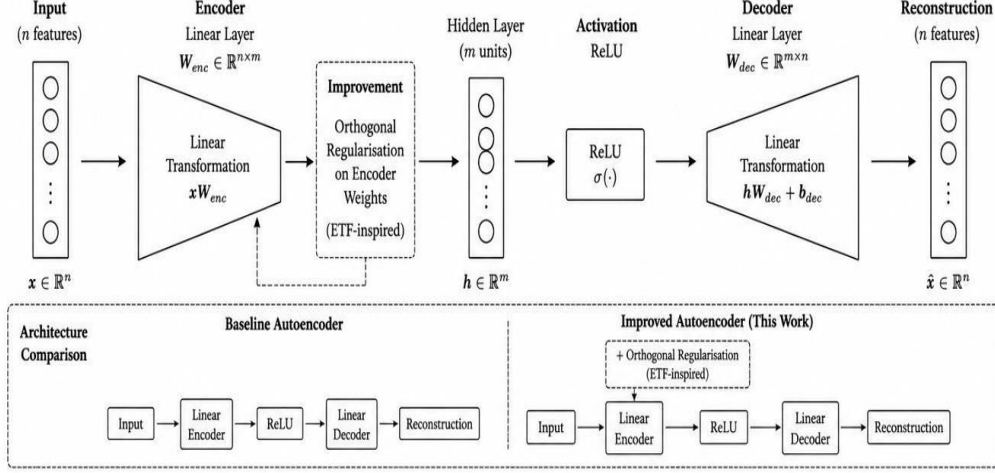


Figure 3: Architecture of the Model with Proposed Improvement

The regularisation penalty was chosen as the mean square of the off diagonal elements and in the optimisation process, the reconstruction loss was minimised along with the regularisation penalty. Therefore, it was desired to teach the model to learn more disentangled and diverse representations while retaining the reconstruction accuracy.

3.6 Training Procedure

The AdamW optimiser was used for training the models since it offers a more stable optimisation process and enhances generalisation by separating weight decay from gradient updates. The learning rate is crucial and so it has been decreased based on CosineAnnealingLR scheduler which slowly reduces the learning rate in the various epochs which is advantageous to achieve quick convergence in early epochs and fine optimisation in later epochs. To avoid exploding gradients, and to increase the numerical stability (maximum norm = 1.0), gradient clipping is used. To speed up the GPU computation and lower the memory usage, Automatic Mixed Precision (AMP) was used without impacting the accuracy of the models [9]. To check the generalisation ability, validation loss was measured during the training process, and to prevent useless computation, early stopping was used, that is, training was stopped when the validation loss stopped decreasing. Lastly, checkpointing saved the best performing model parameters and the next evaluation was done using the most optimal model that was achieved during training.

3.7 Experimental Configuration

The experimental set-up was designed to provide a suitable numerical efficiency, repetition and fair test of the proposed model. The learning rate was chosen to be 0.001 to ensure stable convergence without oscillations in the optimization process. Optimized batch size was 2048 which maximized utilization of the GPU and reduced the variance of the gradients. 500 epochs were set but training was stopped early by early stopping when it was met so that unnecessary

calculations were avoided. The weights were decayed exactly at 0.0 (strong superposition) and 0.1 (weak superposition) for investigating the various regimes of representation. The results indicate that the orthogonal regularisation coefficient ($\lambda = 0.05$) provided a good compromise between accuracy of reconstruction and disentanglement of features. To explore variations of the neural scaling behaviour between different types of models of varying representational capacities, hidden dimensions of 16, 32 and 64 were chosen in each experiment, which were repeated 3 times to test the statistical reliability of the result.

Table 1: Configuration Summary

<i>Parameter</i>	<i>Value</i>	<i>Justification</i>
Learning Rate	0.001	Provides stable optimisation and reliable convergence.
Batch Size	2048	Improves computational efficiency and stabilises gradient estimation.
Maximum Epochs	500	Allows sufficient training while early stopping prevents overtraining.
Weight Decay	0.0 / 0.1	Enables comparison between strong and weak superposition regimes.
Orthogonal Regularisation (λ)	0.05	Encourages feature orthogonality without dominating reconstruction loss.
Experimental Runs	3	Supports statistical analysis and reduces randomness.
Hidden Dimensions	16, 32, 64	Evaluates neural scaling behaviour across increasing model capacities.

3.8 Evaluation Metrics

Evaluation of the proposed model was done with representation learning specific metrics and not with classification. To assess the reconstruction quality and generalisation, Validation Loss (Mean Squared Error) was used which indicates its reconstruction capability of the features with smaller value representing more accurate reconstruction of features. The fraction of encoder feature vectors with unit norm, measured by Representation Fraction, showed the good use of the latent capacity of the model. The off-diagonal values of the encoder Gram Matrix were evaluated using Mean Squared Overlap (MSO), showing good orthogonality of features and less superposition of feature representations with lower value. A power law relationship between the validation loss and the hidden dimension was used to assess neural scaling behaviour, and was called the Scaling Exponent [10]. Computational efficiency was tested using Training Time and model complexity and scalability under various hidden dimensions is measured with Parameter Count. Finally, experiments were repeated and an independent two-sample t-test was conducted to determine the statistical significance of the performance difference between the baseline and improved models.

4. Experiments and Results

The basic superposition autoencoder (SAE) and the proposed orthogonally regularised model (OR-SAE) with both hidden dimensions of 16, 32 and 64 were used for experimental assessment. The validation loss, representation fraction, mean squared overlap, statistical significance, computational cost and scaling behaviour were used to evaluate the performance, offering a thorough evaluation of the reconstruction quality, representation efficiency and scalability.

Both models exhibited a neural scaling behaviour with a regular decrease of the validation loss as the model size increased (Figure 4). The scaling exponent ~ 0.49 was estimated and they were able to reproduce the power-law scaling behaviour in the original paper. Though the validation losses of the improved models were a bit higher than that of the baseline, there were no significant differences between the baseline and the improved models in any model size, and the validation results showed that the proposed orthogonal constraint did not have an impact on the reconstruction performance, but mainly provided representational structure.

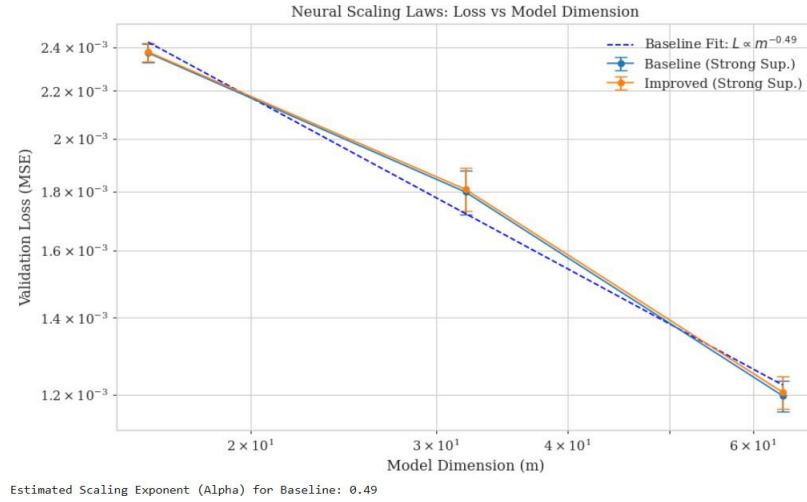


Figure 4: Loss vs Model Dimension

This advantage of the proposed improvement is shown in Fig. 5 where the correlation of capacity of representation and feature overlap are considered. For both the models, the number of latent features that were active were equal to 1.0, representing fraction. But when they used orthogonal regularisation, it facilitated more disentangled/representative feature representations, without sacrificing the representational capacity, as shown by the mean squared overlap being reduced from ~ 0.295 (regularisation) to 2.72×10^{-5} (orthogonal regularisation).

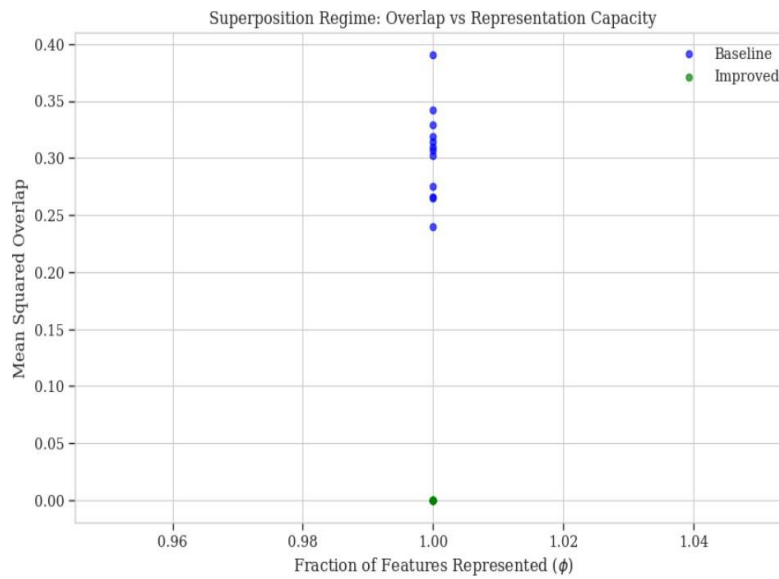


Figure 5: Overlap vs Representation Capacity

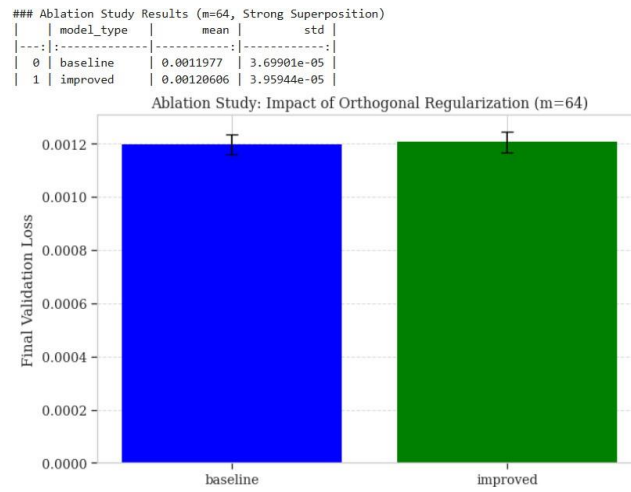


Figure 6: Ablation Study Result

Ablation study (Fig. 6) additionally revealed that reconstruction loss might be a little better without the orthogonal regularisation, but the influence was weak (thirteen or so 0.70%). Similarly, the quantitative evaluation (Figure 7) showed that the feature overlap was maintained low with the improved model and the reconstruction accuracy was found to be comparable for all the hidden dimensions.

Quantitative Evaluation Table

model_type	hidden_dim	Avg_Loss	Std_Loss	Avg_Overlap	Avg_Rep_Frac	Avg_Time
baseline	16	0.00237257	4.37241e-05	0.353899	1	19.978
baseline	32	0.00179799	7.91743e-05	0.314313	1	19.9142
baseline	64	0.0011977	3.69901e-05	0.294765	1	19.4658
improved	16	0.00237708	4.41519e-05	1.93789e-06	1	23.8923
improved	32	0.00180765	7.76787e-05	8.99587e-06	1	24.0785
improved	64	0.00120606	3.95944e-05	2.71562e-05	1	23.8389

Figure 7: Quantitative Evaluation

Statistical Test (m=64)
T-statistic: -0.2675
P-value: 8.0234e-01
Result: Not Statistically Significant (p >= 0.05)

Figure 8: Statistical Test Result

The p value for the result of the statistical analysis (Figure 8) was 0.802 indicating that there was no statistically significant difference between the validation loss of the two models. From

this, it can be concluded that it is not possible to confidently conclude that the slightly higher reconstruction error is due to the proposed modification.

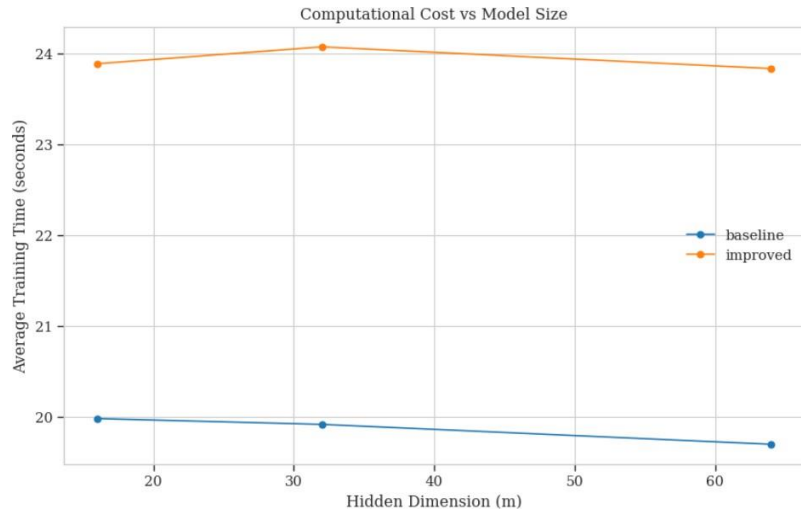


Figure 9: Computational Cost vs Model Size

Computational analysis (Figure 9) revealed that training time increased only slightly because of the extra calculations of orthogonality, and no increase in the number of parameters. Finally, the overall comparison (Figure 10) shows that despite a minor reconstruction error, the proposed model can gain much higher representation quality and consequently learn much more orthogonal and interpretable representations while maintaining the neural scaling property of baseline.

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### Final Model Comparison (m=64)
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model_type	Loss_Mean	Loss_Std	Overlap_Mean
baseline	0.0011977	3.69901e-05	0.294765
improved	0.00120606	3.95944e-05	2.71562e-05

Percentage Improvement in Loss: 0.70%

Figure 10: Final Model Comparison

5. Critical Discussion

Experimental results are provided that show the implemented framework is able to successfully replicate the neural scaling behaviour described in the selected NeurIPS paper. The validation loss monotonically decreased as a function of the hidden dimension with a power-law exponent that was estimated to be 0.49. This is consistent with the theoretical argument that the higher representational capacity of the model, the better it can represent sparse features leading to a lower reconstruction error.

The results showed that the proposed orthogonal regularisation was able to efficiently decrease the feature overlapping without significantly affecting the reconstruction performance. The number of order of magnitudes the MSO could drop while still having a full representation capacity was quite large and this gave more independence to the order of the weight vectors of the encoder [11]. While there was a slight rise of about 0.70 % in the validation loss, the difference could not be proven to be statistically significant ($p = 0.802$). Hence, it is possible to improve the efficiency of representation with no significant degradation in reconstruction quality by the proposed regularisation.

The experimental results are mostly consistent with the results presented in the NeurIPS paper. Both studies demonstrate that higher latent dimensionality always leads to better performance of reconstruction, and the neural scaling robustness and the performance of reconstruction are plausibly explained by representation superposition. But, unlike the original paper where the main emphasis was on the analysis of the scaling behaviour in different weight decay regimes, this work added an additional orthogonality constraint that would promote feature separation [12]. In the original study, the practical implementation has been done only on the effect of implementing regularised representations on the results of overlap and reconstruction – the practical implementation of the proposed scheme was not performed. The small size of the synthetic dataset, the simplified architecture of the autoencoder and the smaller amount of computing resources used in this experiment than in the original experiments account for the slight differences in the scaling behaviour.

The theoretical framework was adequately reproduced but there are still some practical limitations. The toy model considers idealised sparse feature distributions which are not sufficient to represent the complexity of real-world data. Furthermore, synthetic datasets lack in terms of feature interaction and experiments were conducted in somewhat limited dimensions of the hidden space and number of repeats because of limited computational resources [13]. However, the results are not generalisable to larger neural networks and foundation models due to these limitations.

Future work will be focused on methods of adaptive orthogonal regularisation which dynamically change the strength of regularisation during training. More empirical validation would be enlarging feature spaces, having more complicated sparse autoencoders, and real-world datasets. Additionally, the use of these techniques in transformer-based models and LLM architectures could help assess the effectiveness of efficient representations for scaling, interpretability, and performance in real-world deep learning applications.

6. Ethics and Scalability

The proposed implementation enables the interpretability by orthogonal latent representations, which leads to less redundant and more interpretable learned features than when highly overlapping learned features are used. This increased transparency can aid in the more general explainable AI objectives and in the decrease of ambiguity in feature learning. However, the field of neural scaling is still emerging and there is the possibility to use such a large-scale model with increasing capacities that are easily deployable and low cost for responsible deployment [14].

The model was moderately memory consuming and training time did not significantly increase with the added orthogonality calculations, so that it can be applied to larger experiments. Additional improvements are possible by taking advantage of distributed training and parallel processing in future large-scale implementations. Particular attention must be paid to energy-saving training methods, continuous monitoring of deployment models, compression technologies to make effective use of resources, and even open science to make them repeatable and transparent and to enable them to be independently verified.

7. Conclusion

With this project, we have been able to replicate and extend the framework we selected at NeurIPS to investigate the theoretical relationship of representation superposition and scalability of neural networks. The baseline superposition autoencoder was built and improved utilizing an orthogonal regularisation technique to make the features orthogonal to each other and keep the capacity to reconstruct the input. The experimental results showed that the neural networks had a good scaling behaviour, and its representation efficiency was significantly improved without increasing the reconstruction loss. The project offers a reproducible implementation, extensive experimental analysis and critical analysis of orthogonal feature learning. In future work, the proposed approach can be further investigated in applicability of other feature spaces, real-world datasets, sparse autoencoders, and also in transformer-based architectures.

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Code Artefact

GitHub Repository: <https://github.com/mohamedyounus1999-lab/Deep-learning-GEN-AI-.git>